

Pop Quiz) Take gen step on BS of 4-tokens
 on 30B param dense model on TPU v5e 4x4
 slice in int8 w/ bf16 FLOPs, 8192 context,
 100kB/token KV caches

$$i) T_{\min, \text{step}} = \frac{4 \cdot 100 \cdot 10^3 \cdot 8192}{4 \cdot 4 \cdot 8 \cdot 2 \cdot 10^{11}} + \max \left(\underbrace{\frac{2 \cdot 4 \cdot 30 \cdot 10^9}{4 \cdot 4 \cdot 1.97 \cdot 10^{14}}}_{7.6 \cdot 10^{-5} \text{ s}}, \underbrace{\frac{30 \cdot 10^9}{4 \cdot 4 \cdot 8 \cdot 2 \cdot 10^{11}}}_{2.2 \text{ ms}} \right)$$

$\approx 2.5 \text{ ms}$

MLPs are comm-bounded

we could also see this since our total token BS
 is 4 per fw pass $\ll B_{\text{crit}} = 240$.

ii) $B = 256 > B_{\text{crit}} = 240 \Rightarrow$ MLPs are comp bound
 For thoroughness:

$$T_{\min, \text{step}} = \frac{256 \cdot 100 \cdot 10^3 \cdot 8096}{16 \cdot 8 \cdot 2 \cdot 10^{11}} + \max \left(\underbrace{\frac{2 \cdot 256 \cdot 30 \cdot 10^9}{16 \cdot 1.97 \cdot 10^{14}}}_{\approx 4.9 \text{ ms}}, \underbrace{\frac{30 \cdot 10^9}{16 \cdot 8 \cdot 2 \cdot 10^{11}}}_{\approx 4.6 \text{ ms}} \right)$$

$\approx 16 \text{ ms}$

$= 20.9 \text{ ms}$

Q1) i) Per layer:

$$\text{Attn: } 2DNH + 2DKH + D \ll 2DNH + 2DKH$$

MLP: 3DF

$$\begin{aligned}\text{Total} &= L(2DNH + 2DKH + 3DF) + DV \\ &= 64 \cdot (2 \cdot 4096 \cdot 32 \cdot 256 + 2 \cdot 4096 \cdot 8 \cdot 256 + 3 \cdot 4096 \cdot 16384) \\ &\quad + 4096 \cdot 32128 \approx 16.4 \text{ GB params}\end{aligned}$$

$$\begin{aligned}\text{ii) KV cache size/token} &= 2 \cdot H \cdot K \cdot L \\ &= 2 \cdot 256 \cdot 8 \cdot 64 = 262144 \text{ bytes} \\ &\approx 262 \text{ KB}\end{aligned}$$

Q2) Serve on TPU v6e 4x4 slice.

int8, $T = 128 \text{K}$

$$\begin{aligned}\text{i) KV Cache Size/Seq} &= 2HKLT = 2 \cdot 256 \cdot 8 \cdot 64 \cdot 128 \cdot 10^3 \\ &\approx 3.4 \cdot 10^{10} \text{ bytes} = 34 \text{ GB}\end{aligned}$$

$$\text{Model size} = 18.4 \cdot 10^9 \text{ bytes} = 18.4 \text{ GB}$$

$$\text{Total mem} = 16 \cdot 16 \text{ GB} = 256 \text{ GB}$$

$$BS_{\max} = \frac{256 - 18.4}{34} \approx 7$$

ii) Dropping $K=1$ scales down KV cache size/token by $\frac{1}{8} \Rightarrow BS_{\max} = 8 \cdot 7 = 56$

Q3)
$$\frac{\text{Model Size}}{(\# \text{ devices}) \cdot W_{\text{HBM}}} = \frac{18.4 \cdot 10^9}{16 \cdot 8.2 \cdot 10^{11}} \approx 1 \text{ ms}$$

assuming full utilization of HBM

Q4) what does ICI look like on a 4x4?

$\Rightarrow$ 2 hardware axes, doubled bus

Roofline bound on TP?

$$\Rightarrow 2 \cdot F / \left(\frac{C}{W_{\text{ci}}} \right) = 2 \cdot 16384 \cdot \frac{9.0 \cdot 10^{10}}{3.94 \cdot 10^{14}} = 7 < 4 = 16$$

We are comm-bound with TP only.

Prefill:

From our TP roofline, if we use the whole slice for TP, we will be comm-bounded. What I would do is use 8-way TP which will still be slightly comm-bounded but right on the edge with 2-way sequence parallelism.

Generation:

For generation, our only option is model parallelism. Per layer, since a majority of the parameters are within the mlp (approximating by ignoring einsum):

$$T_{hbm\ comm} = \frac{2DF}{16 \cdot W_{hbm}}$$

$$T_{ici\ comm} = \frac{2BD}{W_{ici}}$$

Want:

$$T_{ici\ comm} < T_{hbm\ comm} \Rightarrow \frac{2BD}{W_{ici}} < \frac{2DF}{16 \cdot W_{hbm}}$$

$$B < \frac{F \cdot W_{ici}}{16 \cdot W_{hbm}} = \frac{16384 \cdot 9.0 \cdot 10^{10}}{16 \cdot 8.2 \cdot 10^{11}}$$

$\Rightarrow B < 112$ which will probably be true in practice since 112 is a pretty big batch size of requests to arrive at once.

For generation, I'd use 16-way TP since we're already comm-bounded and as shown above at reasonable inference batch sizes our ici comm time will be less than the time to load from HBM $\rightarrow$ MXU. Since K (# of KV heads) is < 16 but divides it, I'd shard the KV cache along the head and batch dimensions.

Under this sharding each chip would store

$\frac{1}{16}$ of all the parameters so assuming $BS=1$:

$$\text{Step Time}_{\min} = \frac{(34 + 18.4) \cdot 10^9}{16 \cdot 8.2 \cdot 10^{11}} \approx 3 \text{ms}$$

This comes from the fact that we're still bottlenecked by loading params from HBM and each device has roughly ($\frac{1}{16}$) of the data.

Q5) $E=16$ $K=2$

i) Per layer:

$$\text{Attn} = 2DNH + 2DKH$$

$$\text{MLP}_{\text{total}} = 3DF \cdot E$$

$$\text{MLP}_{\text{act}} = 3DF \cdot K$$

$$\text{Total} = 2DV + 2 \cdot (2DNH + 2DKH + 3DFE)$$

$$= 2 \cdot 4096 \cdot 32128 + 64 \cdot (2 \cdot 4096 \cdot 32 \cdot 256 + 2 \cdot 4096 \cdot 8 \cdot 256 + 3 \cdot 4096 \cdot 16384 \cdot 16)$$

$$\approx 211B$$

$$\text{Activated} = 2DV + 2 \cdot (2DNH + 2DKH + 3DFK)$$

$$\approx 31B$$

ii) Since MLP accounts for most of the FLOPs, we see the loss on a single device where

become comm bound for one of the matrices

(all 3 are the same size). Also assuming

we're in generation and our sequence dim is trivial.

$$AI = \frac{2BDFK}{2BD + 2DFE + 2BFK} \approx \frac{2BDFK}{2DFE} = B \cdot \frac{K}{E}$$

$$B \cdot \frac{K}{E} \geq \text{Intensity (vbe)} = 240$$

$$\Rightarrow B \geq \frac{240 \cdot E}{K} = 1920$$

iii) The KV Cache size per token stays the same as 262 KB since MoE doesn't change the K.V projections.

iv) Again, our linear projections dominate FLOPS so:

$$\begin{aligned} \text{FLOPS} &\approx 2 \cdot (\text{activated params}) \cdot T = 2 \cdot 31 \cdot 10^9 \cdot T \\ &= 62 \cdot 10^9 \cdot T \end{aligned}$$

Q6) Each FFW weight shared as $[E_z, D_x, F_y]$

x only used during train as FSDP dimm

i) TPU v5e 8x16 slice, $Y=8$, $Z=16$, int 8

$$T_{\text{weights}} \approx \frac{211 \cdot 10^9}{8 \cdot 16 \cdot W_{\text{hidden}}} \quad (\text{most parameters are in the MLPs so this is a good})$$

$$\approx 2ms$$

approximation)

$$\text{Per chip mem used} = \left(\frac{211 \cdot 10^9}{8 \cdot 16} \right) \approx 1.65 \text{ GB}$$

$$\text{Remaining per chip mem} = 16 - 1.65 = 14.35 \text{ GB}$$

ii) Again, assuming weights are stored in int8:

$$N_{\min} = \frac{211}{16} = 14 \text{ vbes} \Rightarrow 7 \times 2 \text{ slice}$$

(no wraparound connections though)

Q7) For 2D sharding:

$$T_{\text{math}} = \left(\frac{2BDF}{xyz} + \frac{2BDF}{xyz} \right) = \frac{4BDF}{Cxyz}$$

Assuming bf 16, ^Cbidirectional icc:

$$T_{\text{comm}} = \frac{2B \cdot \left(\frac{D}{x} \right)}{2 \cdot W_{\text{ic}}} + \frac{2 \cdot 2B \cdot \left(\frac{F}{yz} \right)}{W_{\text{ic}}} + \frac{2B \cdot \left(\frac{D}{x} \right)}{2 \cdot W_{\text{ic}}}$$

$$= \frac{4BD}{2xW_{\text{ic}}} + \frac{4BF}{yzW_{\text{ic}}}$$

Since we are free to set the topology, we want to minimize x, yz
only the product matters for T_{comm}

$$yz = N/x, \quad x = N/yz$$

$$T_{\text{comm}} = \frac{4BD}{2xW_{\text{ic}}} + \frac{4BFx}{NW_{\text{ic}}}$$

$$\frac{\partial T_{\text{comm}}}{\partial x} = \frac{-4BD}{2w_{ic}x^2} + \frac{4BF}{Nw_{ic}} = 0$$

$$\Rightarrow \frac{BF}{Nw_{ic}} = \frac{BD}{2w_{ic}x^2} \Rightarrow x^2 = \frac{D}{2w_{ic}} \cdot \frac{Nw_{ic}}{F} = \frac{ND}{2F}$$

$$x_{\text{opt}} = \sqrt{\frac{ND}{2F}}$$

$$V_z = N/k = \sqrt{N^2 \cdot \frac{2F}{ND}} = \sqrt{2FN/D}$$

Assuming $F = 4D$ like in specified model:

$$x_{\text{opt}} = \sqrt{\frac{ND}{2 \cdot 4D}} = \sqrt{\frac{N}{8}}$$

$$T_{\text{comm}} = \frac{4BD}{2 \cdot \sqrt{\frac{N}{8}} \cdot w_{ic}} + \frac{4BF \cdot \sqrt{\frac{N}{8}}}{N \cdot w_{ic}}$$

$$= \frac{2BD}{w_{ic} \cdot \sqrt{\frac{N}{8}}} + \frac{16BD \cdot \sqrt{\frac{N}{8}}}{N \cdot w_{ic}} = \frac{2BD}{w_{ic}} \cdot \left(\sqrt{\frac{8}{N}} + 8 \cdot \sqrt{\frac{N}{8N^2}} \right)$$

$$= \frac{2BD}{w_{ic}} \left(\sqrt{\frac{8}{N}} + \sqrt{\frac{64N}{8N^2}} \right) = \frac{4BD}{w_{ic}} \cdot \sqrt{\frac{8}{N}}$$

$$= \frac{\sqrt{128} \cdot BD}{w_{ic} \cdot \sqrt{N}}$$

For normal 3D model parallelism ($[D, F_{xyz}]$):

$$T_{\text{math}} = \frac{2BDF}{xyz \cdot C} \cdot 2 = \frac{4BDF}{xyz \cdot C}$$

$$T_{\text{comm}} = \frac{2 \cdot 2 \cdot BD}{3 \cdot w_{ic}} = \frac{4BD}{3w_{ic}}$$

The time for math comp is the same so we're looking for when:

$$T_{\text{comm}, 3D} > T_{\text{comm}, 2D} \Rightarrow \frac{4815}{3w_{ic_i}} > \frac{\sqrt{128} \cdot 4815}{w_{ic_i} \cdot \sqrt{N}}$$

$$\Rightarrow \sqrt{N} > \frac{3 \cdot \sqrt{128}}{4} \Rightarrow N > \frac{9 \cdot 128}{16} = 72$$

When we have more than 72 chips in our slice (assuming slice is chosen s.t. we have bidirectional w_{ic_i} over all axes), the 2D scheme will outperform the 3D one.